

Experiment No. 10

Title

Missionaries and Cannibals Problem Using Artificial Intelligence

Aim

To develop a Python program that solves the Missionaries and Cannibals Problem using Artificial Intelligence.

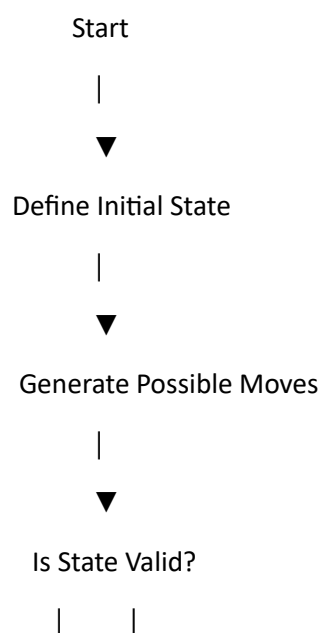
Objective

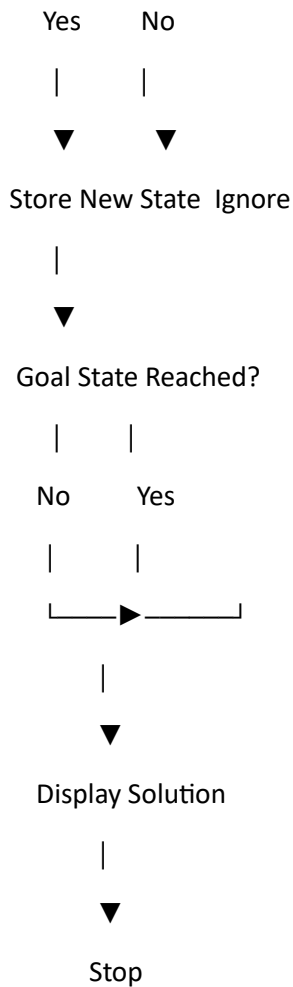
The objective of this experiment is to understand state-space search by solving the Missionaries and Cannibals Problem without violating the constraint that cannibals should never outnumber missionaries on either river bank.

Algorithm

- Step 1:** Define the initial state and the goal state.
 - Step 2:** Check whether the current state is valid.
 - Step 3:** Generate all possible boat movements.
 - Step 4:** Move missionaries and cannibals across the river.
 - Step 5:** Avoid revisiting invalid states.
 - Step 6:** Repeat until the goal state is reached.
 - Step 7:** Display the solution.
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Flowchart





Python Program

```
from collections import deque
```

```
start = (3, 3, 1)
```

```
goal = (0, 0, 0)
```

```
moves = [(1,0),(2,0),(0,1),(0,2),(1,1)]
```

```
def valid(m, c):
```

```
    return (m == 0 or m >= c) and ((3 - m) == 0 or (3 - m) >= (3 - c))
```

```
queue = deque([(start, [])])
```

```
visited = set()
```

```
while queue:
    (m, c, boat), path = queue.popleft()

    if (m, c, boat) == goal:
        print("Solution:")
        for state in path + [(m, c, boat)]:
            print(state)
        break

    if (m, c, boat) in visited:
        continue

    visited.add((m, c, boat))

    for dm, dc in moves:
        if boat == 1:
            nm, nc = m - dm, c - dc
            nb = 0
        else:
            nm, nc = m + dm, c + dc
            nb = 1

        if 0 <= nm <= 3 and 0 <= nc <= 3 and valid(nm, nc):
            queue.append(((nm, nc, nb), path + [(m, c, boat)]))
```

Sample Output

Solution:

(3, 3, 1)

(3, 1, 0)

(3, 2, 1)

(3, 0, 0)

...

(0, 0, 0)

Result

The Missionaries and Cannibals Problem was successfully solved by reaching the goal state without violating the problem constraints.

Conclusion

Thus, the Missionaries and Cannibals Problem was successfully implemented using state-space search, ensuring that missionaries were never outnumbered by cannibals on either river bank.